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Artificial Intelligence, Cybercities and Technosocieties

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Abstract Information technologies have made possible the rising of new forms of communities, cities and societies. These changes are analyzed from the perspective of innovation studies, as technological but also social innovations. Starting from the contributions of Ortega y Gasset to the philosophy of technology, and applying these ideas to the information and communications technologies (ICT) system, this article introduces the notions of technosocieties and cybercities. Our aim is to deeply examine the Telepolis project; a digital and global city supported by ICT and artificial intelligence (AI). We pay attention to the different challenges that AI will have to face in upcoming years in technosocieties and cybercities. In our opinion, the future of AI is tightly related with the technological support of this kind of new city and their cybercitizens. Finally, we claim that there won't be a shared public space in the infosphere till public organizations acknowledge the importance of promoting and maintaining this new and already needed digital agora.

Keywords Innovation studies · Social innovation · Infosphere · Telepolis · Ortega y Gasset · ICT

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1 Technology and Supernature in Ortega y Gasset

In his book "Meditation on Technology", published in 1939, Ortega y Gasset claimed that "man, by means of his gift of technology, segregates a supernature adapting nature to his needs" (Ortega 2015, 65). He claims a central idea in his philosophy of technology: "technology is the opposite of the adaptation of the subject to the medium as it is the adaptation of the medium to the subject" (Ibid, 65–66). He had previously defined technology as "the reform that man imposes on nature with a view to satisfying his needs" (Ibid, 63). He also added: "Technology, then, is the energetic reaction against nature or circumstance that leads to the creation, between nature and man, of a new nature above nature; a supernature" (Ibid, 64). 80 years have passed since Ortega published these ideas which made him a pioneer in the philosophy of technology although he is hardly ever cited internationally. To our understanding, his proposals deserve to be revisited and reinterpreted. In this section, we will reexamine it in order to analyze some of the social consequences of present-day information and communications technologies (ICTs, from now on). Some ICTs continue to transform nature. For example, when they converge with nanotechnologies and biotechnologies and contribute to generating new materials like grapheme (nanomaterials) or synthetic corn (genetically modified organisms). If we take the expression "artificial intelligence" in a strict sense, the convergence between ICTs and the neurosciences deserves special attention. This merging can actually modify cognitive processes and even generate *artificially modified human intelligences*. However, we will not consider these "supernatures" but we will pay attention to the ICTs that are designed to implement and modify social relations and interactions. Following a general philosophical excursus (Sect. 2) in which we will clarify our own analytic perspective, we will apply Ortega's ideas (Sect. 3) to the conceptual analysis of today's techno-cities and techno-societies. We will see how information societies (Castells 1996) superimpose themselves on previously existing cities and societies. Ortega did not really believe in the concept of nature but he invented the word "super-nature" to refer to the different ways of artificializing the biosphere. In fact, in a certain passage of this book he even said that "nature means nothing more here than that which surrounds man, circumstance" (Ibid, 60). We will adopt this Orteguian perspectivism throughout this text and we will refer further in the following section. We also accept other of the central ideas of Ortega's philosophy:

> I am myself and my circumstance, and only if my circumstance is saved, will I be saved (Ortega, *Meditaciones del Quijote*, 1914, OC, I, 322).

We also point that the exterior circumstance in which human beings live has radically changed since the end of the 20th century and it will continue to be transformed by different ICT innovations. One of the latest examples of these pervasive technologies are social media channels (Facebook, Twitter, etc.) that have delivered different social innovations that result in the emergence of *technosocieties* and *cybercities*. Paraphrasing Ortega once again, we will say that ICTs have made the appearance of a *techno-circumstance* possible (Echeverría 2013) and defining this as everything around us that works based on the ICT technological system. In recent decades, this techno-circumstance (or cyber-circumstance) has been developed significantly. ICT artifacts (digital radios and televisions, computers, cellphones, tablets, videogame consoles, credit cards, bar codes, etc.) proliferate all around us, and they have considerably modified human relations despite people also continue to relate and interact in the old ways. On the other hand, we must underline that the great majority of these ICT artifacts are completely new or have profoundly modified previously existing apparatuses such as old televisions and analog telephones. Therefore, a whole set of *disruptive technological innovations* have been introduced in society. This is one of the main reasons to analyze them from the perspective of innovation studies and even from the perspective of a possible *philosophy of innovation*. The most common approach is to focus on product innovations. Something that is usual in ICTs with a bunch of new digital and information objects. However, in this text we will deal with what we could call *relational ICTs*; *the ICTs that generate new forms of social relations and interaction*. What lies here is whether these relational ICTs are able to generate complex systems of relations and interactions that could be described as *cyber-cities* and *techno-societies*. If this is the case, we could talk about *cyber-urban* and *techno-social disruptive innovations.* Ortega also claimed:

> A man without technology, with no reaction to the medium, is not a man (Ortega, *o.c.*, 66).

This idea can be glossed in this way; man in a natural state is not a man, he is simply a monkey (which, we must not forget, is also a social animal). The tendency to transform the environment using different technologies is a constant in the processes for humanizing nature. This is why it is possible to distinguish between natural environments (without direct human intervention) and artificial environments. However, the frontier between natural and artificial is diffuse. And in some cases, there has been a high degree of hybridization between nature and technology. Ortega went so far as to describe human beings as *ontological centaurs*. A very interesting idea which we have reinterpreted by speaking of *techno-bodies* and *techno-agents*:

> The being of man possesses the strange condition that it is partly close to nature, but partly not. That it is, simultaneously, natural and extra-natural, a kind of ontological centaur (Ibid., p. 82).

To this statement we would like also to add another characteristic of this technological capacity of human beings: its *recursiveness*. Human beings transform nature. At first, they build adobe and brick houses but after that, they transform this first degree of “techno-nature” (the Greek *oikos* and the Roman *domos*) generating more complex and more artificial entities such as cities (*polis*, *urbis)*. This process continues along the times and centuries later, the metropolis and the megalopolis have risen. But they also transform the old hamlets and villages, turning them into great cities or connecting them by roads, wagons, ports, ships, highways, cars, train tracks, trains, airports, airplanes and currently space stations, telecommunications satellites and “information highways” as Al Gore called telematic networks like the

Internet. The recursiveness of *homo technologicus*' actions involves using many and very different technological systems when it comes to building cities, cyber-cities and technosocieties. Some of these systems continue to be farming, livestock, mining and traditional systems but also industrial technologies. What is important here is that *some technological systems superimpose themselves upon others and are assembled together with them*, generating at the same time *techno-human centaurs* in the end, in which body and technology are more or less hybridized (Actor-Network Theory from Bruno Latour).

In the case of ICTs human beings not only transform the biosphere but also the artificial environments (villages, cities) that humanity had previously created. According to Ortega, this technological impulse of human beings arises from the fact that no system, either artificial or natural, fully satisfies human beings' needs and desires. There is a reason for this: *new desires and new needs always appear.*

> Human needs are objectively superfluous and only become needs for the person who needs wellbeing and for the person for whom living is, essentially, living well (Ibid., 69). The key value for technology is, then, wellbeing: "man, technology, and wellbeing are, ultimately, synonymous" (Ibid.).

On this point, we disagree slightly from Ortega. He is partly right but human technological actions are also governed by other values not only wellbeing. Other motivations like the control of the environment in favor of utility, efficiency and efficacy are just other typically technological values. Independently of this axiological debate, we would like to underline Ortega's central contribution: the *recursive superimposition of some technological systems over others*. The artificial cities and societies to which we will pay attention are not only super-nature forms. They are superimposed on previously existing cities and societies and they generate social and urban entities with a qualitatively higher degree of artificiality. This is why they could be described as "super-cities" and "super-societies" forcing Ortega's terminology. Although we would like to propose another term coined in our previous works. This is the idea of Telepolis; the possible global city sustained by the ICT system proposed in 1992 (Echeverría 1994) and to which we will allude later. We will summarize this reinterpretation of the Ortegan philosophy of technology by saying that human beings transform the environment recursively. This is something characteristic of artificial intelligence (using this expression here in its technical meaning). Assuming that at some time human beings inhabited an exclusively natural environment and remembering that they were not human but simians, the fact is that they transformed this natural environment and artificialized it in a way, resulting in the appearance of new needs. Those needs are not natural but specifically human such as the need to maintain the *domos*. Subsequently, the domestic habitat has continued to be transformed thanks to human effort and to new technological systems and producing a greater degree of artificiality. And this goes on to today's digital and telematic environments where cybercities and technosocieties are emerging. However, saying that this process of artificializing the environment is recursive does not mean that we are stating that it is lineal or deterministic. As Ortega points out (Ortega, *o.c.*, 72), throughout history, there have been many technological inventions that have failed. The first sentence of

Chesbrough's book on open innovation says that "the majority of innovations fail" (Chesbrough 2011, 23). Innovation does not mean be sustainable in space and time and the need for allocating business models to different inventions have become an imperative need (Osterwalder and Pigneur 2010). Many great technological inventions have been discarded by history precisely because the system of human needs and desires has changed and with it, the human circumstance. Human environments are charged with values and projects which is why they cannot be reduced either to nature or to the world of being. It is important to clarify this last point and that is why we will now make a brief *excursus* regarding ontology and axiology before returning to the central subject of this article.

2 Ontology and Axiology of the ICT System: Some Notes

2.1 Perspectivism and Technologies

Information and communications technologies have generated a new dimension of the world which we call the *third environment* (Echeverría 1999). Floridi calls it the infosphere, an expression which we will also use.[1] This digital world shaped by the ICTs is usually an object of research for different disciplines: economics, management, sociology, politics, etc. We would like to emphasize that the third environment has an important military component (Milnet) as well as a financial one (Swift). But his third environment is not limited to the Internet or the World Wide Web. We just browse some shinny and visible parts of this ecosystem but it is much more extensive. What is most visible about Telepolis, the global cybercity, is Internet and social networks. The gateway of each user joint to the identifications of the user (URL, electronic address, credit card number, access codes, etc.) define our *point de vue* (Leibniz) in digital space–time as well as our *new circumstance* (Ortega y Gasset). We will call this the *techno-circumstance* but also the *cyber-circumstance.* This is something controlled by an Internet Service Provider (ISP). So each user with access to the third environment generates his or her own world of life (*Lebenswelt*) there. This is a world of life populated by information and digital objects (Floridi's infosphere) but also by relations with other subjects through the networks. From now on, we will focus on these links among techno-agents because they provide the relational basis for cybercities and technosocieties. We wish to make it clear that we accept *philosophical perspectivism* presented by Leibniz in the *Monadology* and later developed by several authors among them Gabriel Tarde, Friedrich Nietzsche and Ortega himself. In fact, Ortega formulated his own "*doctrine of the point of view*" at the end of the book "El tema de nuestro tiempo" (1923). In this book he also presented his *Estimative* or general science of values which should not be confused with ethics or with moral philosophy, being far broader than these. Ortegan perspectivism has a strong axiological component as he

[1] Floridi defined the infosphere so: "It is a context constituted by the whole system of information objects, including all agents and patients, messages, their attributes and mutual relations" (Floridi 2002, 8).

distinguished between a *world of being* and a *world of value* and therefore, between the ontological and axiological dimensions of each point of view. When it comes to researching the infosphere and the digital world, it is necessary to take into account that each subject whether individual or collective introduces his, her or their own estimations for seeing and doing things in the third environment. This tends to be decisive when it comes to establishing relations there, as we shall see later. Substantialist philosophers tend to criticize perspectivist ontologies, accusing them of relativism. Commenting on Leibniz, Gilles Deleuze stated that "neither in Nietzsche nor in Leibniz does perspectivism mean *to each his own truth*; but rather that *the point of view is a condition for the manifestation of truth*" (Deleuze 2009, 143). As for Ortega, also commenting on Leibniz, he wrote that "the world is a perspective" (J. Ortega y Gasset, *La idea de principio en Leibniz*, 315–316). In the case of the digital dimension of the world, individual subjects and groups organize and maintain their own digital world and they do so because they value this positively. Therefore, the informational world is a lot more than a set of digital objects: it also shapes a way of life. Concretely, a *techno-world* that should be distinguished from the natural and urban environments (the first and second environments) where people also live and relate to one another. It is also important to underline that techno-worlds and techno-societies can be researched on different scales. That is, on the micro-, meso- and macrosocial levels. I.e; a couple who have fallen in love in the third environment and have never physically met establish a new kind of sentimental relationship. The same can be said of other kinds of human relationships. In short, informationalism (Castells 1996) involves a new ontology of the world as Floridi has underlined in several of his works (2004, 2010). But it also generates new relations and new actions (from a distance, online, recursive) which are not feasible in the first or second environments. A concrete example is the virality of a meme[2] throughout social media channels. What is important is the new kind of relationship that is established in the social networks (relational innovations and innovations in processes), more than the new objects generated even if they become "trending topics".

2.2 Relational Ontology and Axiology

The ontological perspective that we take here is a *relationalist* one. Our object of study is not (digital, informational) objects but (digital, informational) relations. Having formulated the hypothesis of the existence of the third environment, there are a minimum of two great philosophical problems to face; the *ontology of the digital world* (objects of the infosphere) and the *axiology of the third environment* which is decisive for the subjects and their relationships and interactions. It is necessary to respond to the question about the beings that are in the third environment (who they are, how they emerge, how they change, etc.). Floridi has done so but because they are artificial entities it is also necessary to ask who programs, develops and disseminates them, who maintains them, who perceives

[2] In the Internet, a meme is a digital object (video, image, text, etc.) that spreads rapidly from person to person via a digital platform (or several platforms), usually in a humorous way.

them, who uses them, and in particular, *which values guide this set of actions*. We want also to stress that axiology and ethics are not the same term. While axiology deals with different kind of values likewise economical, technological, political and aesthetical ones, ethics is focused to moral and religious values. Nevertheless, the domain of ethics is currently expanding to other fields like legal and environment values. The ontology of the third environment must have an object component but it must also analyze the subjective, social, artificial and axiological aspects of what happens there. Therefore, it must accept that the objects of the infosphere are artificial and informational as Floridi has argued. This perspective of reflection is undoubtedly interesting. In this text we pay attention to relational and process ontologies and we do not forget that values underlie human actions, including technological actions. Specifically, we observe the third environment as a relational and process time–space rather than as a set of objects and subjects that are there. We are interested in what happens *between objects* and *between subjects*, in their relations and interactions, which happen in a clearly technological and artificial space–time.

2.3 The Hypothesis of the Third Environment

Now, we would like to follow the assumptions of the Spanish philosopher of technology; Miguel Ángel Quintanilla. We consider the ICTs as a technological system and technological systems to be *systems of human actions*, not only a set of artifacts, mediums or instruments (Quintanilla 1989). We consider that the info-objects that exist are above all, the (ephemeral) results of human actions or *techno-human* actions. Indeed, the subjects of the infosphere are hybrid agents (human and technological), that is, techno-humans.[3] The hypothesis of the third environment states that:

> The ICTs have made possible the emergence of a new social space–time, the third environment, that superimposes itself on the two great space–times that human beings have experienced and that we continue to experience: the physis and the polis (Echeverría 1999).

To develop this hypothesis, we begin with the Leibnizian conception of space and time, according to which they are not substantial entities but relational orders where new objects and new things emerge as well as new ways of relating and interacting. In the end, the consequence of the hypothesis of the third environment is the emergence of a new kind of human agent (I am myself and my circumstance), the *techno-agents* which can be physical (people) but also legal (companies, institutions), as well as purely technological artifacts (servers, robots, automatons, etc.). The precise characterization of this new kind of agency is not easy to produce but that we must be very prudent when it comes to attributing cognitive capacities to these techno-agents, especially intelligence. Do automatons wonder about themselves? Do they try to understand themselves? Do they have intentions, for example,

[3] Bruno Latour's actor/network theory is the reference for this (Latour 2005), even though it has several debatable aspects. The aforementioned Ortegan metaphor, *ontological centaurs*, is much more beautiful.

the intention to understand –*intelligere*? The term *intelligere* has an etymology and there is no reason to give it up: passing Turing's test is not enough to describe a techno-agent as intelligent. Of course, we must not confuse techno-agents with mystic entities, like the supposed trans-humans, either.

2.4 The Technoscientific Revolution

We tend to call the fourth revolution that Floridi talks about the *technoscientific revolution* (Echeverría 2003). On this point, we are also happy to accept Floridi's terminology because we agree with him that Copernicus, Darwin and Freud radically changed our conception of the world. This is something that ICTs have also done in recent decades. However, in this period not only our conception of the world (*Weltanschauung*) has changed but also our own *world of life* (*Lebenswelt*). This fact involves a greater transformation because it affects human relations and interactions. In fact, a good part of our life today unfolds in this new relational space/time which others have called cyberspace, the digital world, electronic space and also the virtual world though we feel that this last designation is open to criticism. We prefer to call it the third environment to underline (in a Leibnizian fashion) that what has changed is *our perspective* which entails (in an Ortegan fashion) that *our circumstance has changed and therefore we ourselves have changed*. The human species has traditionally lived in the country or in cities and this represents the first or the second environment (*physis*, *polis*). Since the broad dissemination of the Internet and other ICTs throughout the globe, our worlds of life have acquired a third dimension; the digital and informational dimension. We live and relate with one another in the third environment despite we already do in the first and second ones. These two earlier forms of world have not disappeared, they are still there. We continue to live and intervene in the first two environments because the third environment superimposes itself upon them and transforms them but does not eliminate or absorb them. The great novelty of the end of the 20th century was the emergence of this new social space–time, the infosphere. This new environment surrounds us daily through multiple screens which are so many windows and entrances to the third environment. As Turkle (1995) wrote previously, today we also live on screens, not only in nature or in cities. Our circumstance has changed radically and with it, our lives; and with them, ourselves. And this has happened because a third dimension of the world has emerged, the informational, electronic and digital dimension.

2.5 Technosocieties

All human societies have been based on technologies starting with prehistoric hunter-gatherer societies. Ortega was right when he said that technological capacity constitutes the human being and therefore, any social form can be described as technosocial; this description is a matter of degree. A brief example; there is no society without language and the use of a language requires certain techniques to learn (simple techniques like modulation and tuning of the voice). However, today's techno-societies are characterized by the fact that technological systems have become an indispensable mediation of reality, to the extent that human relations and actions are only feasible with this mediation. This is what happens in the third environment and this is why we hold that technosocieties as such emerge in this social space where human and social relations are technified to a maximum degree. Coming back to the example of languages, in the first environment languages are oral, in the second are written and in the third are *techno-languages.* Oral expression and hearing, writing, and reading are only possible if the corresponding ICTs exist. These technologies provide of a communication channel that allows people to communicate. Similar idea is behind cultures in the first environment (farming, livestock farming, mining, etc.), in the second (scribes, copyists, cultural trades and professions, different religions, etc.) and in the third. It is a good idea to call informational and digital cultures *techno-cultures* as the corresponding cultural activities are only possible with the mediation of the ICTs. Thus, the use that we propose to make of the prefix *techno-* is systematic. Its use implies that the corresponding noun or verb only has a meaning if some technological subsystem that underlies and sustains the existence of the object or the execution of one action or another is attached to this object or action. Then, cybersocieties (and cybercities) would be the technosocieties whose mediating technological system is an ICT system but with an additional requirement; among the ICTs involved there are some that control the system as a whole. In our case, a technosystem becomes a cybersystem when an ICT system is superimposed upon the previous technological system (i.e. cars), controlling the machine as a whole (onboard computer). The implementation of the ICT can then be of very diverse kinds and relational structures; pyramidal, vertical, transversal or even horizontal (total transparency with no central control). But in order to be a CYBER-ICT, this CYBER component of governance must exist. For example, the robotization of an industrial factory's production turns it into a *cyber-factory* where previously was a techno-factory as the set of productive actions was not manual or traditional but carried out by industrial machines. Once again, we are seeing one technological system superimposing itself upon other previously existing ones. And this is the main reason that could lead us to talk about super-factories in Ortegan terms; one kind of artificialization superimposes itself on another. As the ICTs have generated a new social space–time, which was called *cyberspace*, different technical systems have also generated other techno-spaces and techno-times throughout human history. Some of them have developed and endured while others have degenerated and disappeared. Our proposal regarding technosocieties is an emergentist proposal but it is also an evolutionary proposal. The appearance of cybercities is another phase of the development of technosocieties. In cybercities, the ICT system controls and guides the rest of the technological systems (i.e. Internet of things). This is the reason that we rather to use the prefix *cyber-* to distinguish this from other kinds of techno-city. Cyber-cities superimpose themselves on industrial cities partly to control them. This is why they could also be called super-cities in Ortegan terms.

2.6 Technological and Social Innovations

The emergence of the infosphere or the third environment has meant a disruptive innovation. The ICT technological system has generated *multiple technological innovations,* several of which are breakthrough. Due to this fact, the way of relating and interacting has changed radically and with it, the notion of object (which becomes a techno-object) but also the notion of subject (techno-subjects emerge). In this sense, Artificial intelligence has been decisive as is commonly accepted. This great transformation must certainly be described as revolutionary. We could even say that *a new world* has emerged although we prefer to speak of *a new dimension of the world* that can be summarized in the rising of a new space–time: the third environment. Multiple new objects have appeared in this new environment. The infosphere is and it will continue to be an extensive space for technological innovation. This is the main reason why it is such an important area for innovation studies as well as for the philosophy of technology. But if we take another step and try to do a *philosophy of innovation* (Echeverría 2014, Chap. 5), then it is necessary to distinguish between technological innovations and social innovations. This distinction implies a critical reflection on the dominant paradigm today on innovation studies; the *systemic model* (Nelson 1993; Lundvall 1992). The origins of this approach are based in the recommendations that Vannevar Bush did in his famous report to President Roosevelt (Science, the Endless Frontier, 1945) in favor of a linear model. Its canonical expression is the OECD Oslo Manual (2005) and the European Eurostat. We want to clarify that we are writing this article from a possible philosophy of innovation and not from a philosophy of technology.

3 Cybercities: Telepolis and the Future Challenges of Artificial Intelligence

Human communities have been characterized by sharing something: a territory, an origin, a language, beliefs and a culture or in Ortegan terms; a shared project. This is the motivation that lies behind different communities of practice, organizations and also companies. Societies are much more complex entities because several communities coexist into them and this cohabitation is not free of conflicts between different groups and individuals. From a philosophical point of view, one of the biggest social innovations in history took place in Greece and was called the *polis*. What is most remarkable about it is the existence of a government and some laws to organize social and community coexistence. Plato was the first theoretician of the *polis* in *The Republic*. He noted the importance of the polis as a philosophical task, not as an engineering one and that is why there is no mention about space or buildings which is how we tend to think of present-day cities.[4] In this dialogue, the construction of the polis was strictly philosophical, only conceptual. Its "architects"

[4] In the case of cybercities, there are authors at the end of the 20th century who analyzed its basic structure properly like William Mitchell. His approach was architectural not philosophical and, because of this, he focused on the spaces and constructions on the "superhighway" (infobahn) which was what cyberspace was called then. See Mitchell (1995, 1999).

were Socrates and Adeimantus. When they began the task, the first one said to the second one: “Come then, let us create a city from the beginning in speech” (*The Republic*, 369 c). To do this, they started out with three famous principles: (1) *Interdependence.* “No human being is sufficient unto himself” (Ibid, 369 b). In order to satisfy our basic needs (food, shelter, clothing, etc.) we need the cooperation of other people. A farmer, a bricklayer, a weaver and a shoemaker seemed to them to be the first indispensable occupations. “The indispensable minimum of a city, then, would consist of four or five men” (Ibid, 369d). But as Socrates and Adeimantus thought about the city, the basic needs grew and with them, the actions and skills to satisfy them. Many expertises are necessary in order to provide good conditions for living to many people in a territory. (2) *Plurality of technologies*. Socrates and Adeimantus analyzed the scope of its conceptual *polis* based on the increase in trades and therefore, on the different technical activities that each occupation contributed. Then, a necessary (but not sufficient) condition for the existence of a *poli*s is the plurality of technological systems that make different actions and interactions among people possible. Similarly, this implies the division of work, as Plato underlines, and the existence of markets where each trade exchanges its products with other citizens for satisfying everyone’s needs. Every city has a strong technological component even if the technological systems have evolved and new innovations are present. In Ortegan terms; a city is always a supernature. It is a composition of several technical supernatures which are sometimes hard to fit together causing internal conflicts in the *polis* as well as conflicts with surrounding cities. The warrior’s occupation was necessary to maintain a city according to Plato (Ibid, 374a). (3) *Government.* Warriors and merchants are important powers in any city and society just like priests but the Greek concept of the *polis* is based on the emergence of a singular profession; the politician. His principal task is to govern but he must also propose and enact laws and norms to organize the coexistence of the different occupations and interests. Plato stated decisively that there is no *polis* without governors. This is a metaphor that comes from sailing and refers to governing a ship. A city is made up of several communities of practice (professions) sailing in the same ship (the *polis*) on the ocean of time. As a result, a singular public space appeared in Greek cities; the agora. In this space political debates were held and Socrates, the new philosophers and the sophists all participated actively. Nature continued to be an important subject of philosophical reflection but a new philosophical issue appeared: the city. In addition to the agora, other specific spaces were built also to carry out the work of governing and administering justice. Then, the embryo of the State emerged. The dialectic between country and city (first and second environment, nature and supernature) had appeared in philosophical reflection and has continued up to present times. Having briefly evoked the Greek polis, it is interesting to apply this relational conception to today’s information societies as well as to communities of knowledge (occupations) that make possible this new kind of society. When we refer to Telepolis, we are alluding to the conceptual construction of a new kind of city that grows in the third environment and superimposes itself upon classic cities. Multiple human online and asynchronous interactions take place there and shape the life of Telepolis by means of information and electronic flows. We are not referring to ICTs' impact on how already-existing cities function, turning them into *smart cities*. We are referring to the development of all kinds of activities and professions that are being held in the electronic space and in a digital format. The main difference with past societies is that human relations and actions are produced from the distance and without the usual requirement of traditional societies (or cities) that asks for a gathering of different individuals in a physical place. A city in which human relations and interactions are produced at a distance, online and asynchronously (without being simultaneous) involves a disruptive innovation in *relational ontology*. This fact breaks a fundamental assumption of human coexistence. This coexistence ceases to be based on physical proximity and develops connections by telematics networks. The relational ontology that we argue accepts the existence of informational objects around us (Floridi's infosphere in 1999, 2002) but claims that *informational relations and actions* are more important than objects and things. Paraphrasing Plato, to philosophically create a digital, global cybercity, it is necessary to clarify which crafts and activities are indispensable in it, so that human beings can relate to one another and interrelate at a distance and online. When this will happen, they will exchange the objects produced by their actions (i.e. messages) and to do this, they will have to communicate with one another by means of telematics networks. One of the indispensable professions will be the "networkers". People that design, construct and maintain the telematic networks. Another indispensable one is the programmer whether their languages are oriented toward an object or toward a subject and relation, as occurs in today's social networks. A third trade is linkers or connectors who are usually people but can also be artifacts.[5] Indeed, artificial intelligence is essential for building Telepolis. However, there are other very important trades from the perspective of the cybercitizen such as the *network administrator*. Given the present-day technological structure of the ICT system and until "*intelligent administrators*" appear, a network cannot function without a human administrator. This means without an ultimate human control of the functioning of the network and its sub-networks.[6] "Hardworkers" are also necessary and also suppliers of electricity. Telepolis is a digital city but this does not prevent it from having a material component which must also be produced, maintained, repaired and updated. Information is the raw material that flows in the city and it is produced mainly by the users of the network (Web 2.0 platforms). At the same time, the digitization of documents generated by pre-digital cultures (books, images, sounds, movements, etc.) also contributes to the flow of information. This is the key point in our approach; *the users of the digital networks themselves provide the raw material* and this is why they are the basis of Telepolis cybercitizenship. Then, user's innovation is of outmost importance. As soon as the figure of network administrator appears, *power in the network* also appears. This supervisor has the function of

[5] Castells states that "the power in network society is held by programmers and linkers" (Castells 2011, p. 550).

[6] Of course, the network administrator uses his or her own *hardware* and *software* by means of which he or she controls the actions of the group of users. This entire data can later been stored automatically in "the Cloud" producing another additional trade, carried out by a computer, not by a person, even though this computer had to have been previously designed and programmed for this task.

organizing and coordinating users' relations and interrelations in the network at community and social levels.[7] All kinds of cybercommunities, such as the virtual communities that Howard Rheingold researched, require some kind of administrator and manager. Some communities of practice in the third environment have an open and horizontal structure (i.e. free software communities) while others are closed and vertical. Of course, there are all kinds of intermediate possibilities and hybrid forms. What is important to underline here is that all of these trades and tasks that are at the backbone of these diverse community structures must coexist in a specific sphere of the third environment, usually on a website or in a network, or just in a database. Given a set of networks on a global level, a system of *techno-governance* is necessary, and when appropriate, *cybercontrol*. When Internet began, the first type of government was carried out by the IETF and from the year 2000 onwards, by the ICANN. The governance of the World Wide Web began also to be organized in 1994 with the constitution of the W3C (Alonso 2015, 90) that was a personal initiative of Tim Berners-Lee. The appearance of these "*network super-administrators*" shows that the "trade of governor" is also needed in the third environment. This governance is not only related with intelligence but also with prudence because both of them are of outmost importance for governance. We also recognize prudence as his classical Aristotelian conception (Aubenque 1999). We must do not forget also that these instances of governance are only valid for the Internet. Financial and military networks are not governed by these institutions and have their own systems of control. In civilian networks (Internet) as well as in financial (Swift) and military (Milnet) ones, other instances of citizen governance have followed the first ones and others will undoubtedly follow these initiatives. This is why the challenge of transforming the third environment in a polis arises and this challenging task must be dealt with the aid of artificial intelligence. With the development of the third environment, several information societies with different structures (country, language, etc.) have arisen in this ecosystem. They all share the same technological system; the ICT system. This is what makes it possible to distinguish between societies and technosocieties. Technosocieties are characterized by sharing a technological system, not a territory, or a language or a common origin, not even a culture. What is left to emerge in the third environment is a *res publica*. The most similar thing to the Greek agora is the WWW and that's the reason why it has been so popular and widely known. The growth of the World Wide Web has also fostered a new digital culture that favors the generation of digital commons (Wittel 2014). These digital resources can be produced, used and shared by different users. In this sense is useful to mention significant initiatives of Free/Libre/Open Source Software (FLOSS) like the Wikipedia encyclopedia, Mozilla Firefox browser or HTML5 web standard (Tabarés-Gutiérrez 2015). However, there is still a lot to do before open access to knowledge can be guaranteed for all kind of users. This is something critical for building a public agora in the third environment and therefore, for constituting Telepolis. It seems that a global and digital city is being built supported by electricity and the ICT system. However, its constitution as a

[7] However, this has still not appeared as a political problem because there is no *polis* yet, only communities of knowledge and societies of information (techno-societies).

polis still seems to be distant. Until now, the driver of the development of the ICTs and the third environment has been technological innovation. Perhaps the time has come to promote an organizational and disruptive social innovation which could be the constitution of Telepolis, not just its construction. To achieve this goal, the users should get seriously involved in the government of the networks and particularly, in the democratic endorsement of laws and norms that could rule relations and interactions among ICT users. Consumers and prosumers (Toffler 1981) of information and ICT systems should be aware as their role as cybercitizens and organize themselves as such. From our point of view, this is the social innovation that is emerging today in the third environment; the constitution of cybercitizenship and, when appropriate, the establishment of democratic governance.[8] This makes us wondering if it is possible to build and constitute a digital and global city in the third environment. If this digital global city comes into existence, a lot of intelligence will be needed to govern it. The challenge of AI is to be considered as one of the technological systems that sustain the construction, maintenance, functioning and good government of a possible global city. Artificial intelligence provides important services to the great financial investors and also to intelligence agencies. There is also a civil artificial intelligence which was described well by Floridi when he spoke of information environments and the infosphere. However, there is not yet any democratic artificial intelligence serving the users instead of the network administrators. There are some ongoing initiatives named as "personal assistants[9]" led by some of the great digital players but their role is still unclear. This is one of the great challenges for AI in our times. Based on the hypothesis of the third environment and according to the proposal to build and constitute a global city capable of uniting several information societies and multiple communities of knowledge in the third environment, the future of artificial intelligence will be to provide technological support to this cybercity. It is important to stop thinking of users as consumers of information and start thinking of them as citizens with rights and responsibilities established based on a system of global governance of the new social space-time; Telepolis in the third environment.

3.1 Telepolis and Social Innovation

In order to build a digital and global city, information societies need to keep growing and fostering at the same time communities of knowledge. Many technological innovations are needed and they should be implemented in collaboration with the users. But some deep conceptual changes are also needed in the governance of information societies. Social innovation can carry out an important function in this regard. To magnify the importance of this approach we will deal with three issues:

[8] In this way it is interesting to acknowledge the emergence in last years of of hacktivism. This subversive use of computers and computer networks to promote a political agenda has been clearly shaped by a digital culture.

[9] Siri by Apple, Cortana by Microsoft and Google Now by Google are some examples of how AI is being fueled by these companies. All of them have shown exciting capabilities but these tools are oriented at the end to targeted advertising.

- A philosophy of innovation, in particular, of social innovation.
- An axiology of the third environment based on the outburst of the value "innovation".
- The possible constitution of some kind of *res publica* in the third environment. Or in Floridi's terms, the governance of the infosphere understood as a system of info-subjects, info-relations, and info-actions, rather than a set of information things and objects.

3.2 Towards a Philosophy of Innovation

Innovation is one of the great values of our times but philosophy has hardly paid any attention to it. However, this notion presents important philosophical problems and particularly ontological ones. Innovation is not an object or a thing; it is truly a process. However, few philosophers have paid attention to this category of process, so there is a lot of work to be done. In this sense, it is noteworthy mentioning Alfred Whitehead (1979) but the person who has best set forth the terms of the problem with a kind of orientation toward artificial intelligence is Nicholas Rescher (2007). This is something that has been acknowledged previously (Echeverría 2014, Chap. 5). Secondly, the issue of innovation is very interesting for a relational ontology that focuses on actions and interactions. The notion of novelty is strictly relational, given that *something is new in relation to a previous something*. As the OECD Oslo Manual points out (2005), we have to specify *for whom*, not just who innovates or what is new. This is the institution that articulates the dominant paradigm in innovation studies and according to this Manual an innovation can be "new to the market, new to the world and disruptive innovations" (*Oslo Handbook*, 2005, Sect. 37, 17). The same can be said in the case of social innovations and, in general, for novelties; *something can be new for one person, group or society, but well-known to other people, groups or societies.* This is the main reason why imitation is one of the paths to innovation (Tarde 1903; Godin 2015). Subjectively speaking, a learning process always involves acquiring new knowledge, some of which can be innovative for the people who receive it or invent it. However, Schumpeter made it very clear that not all novelties or inventions are innovations: "innovation is possible without anything we should identify as invention and invention does not necessarily induce innovation but produces of itself no economically relevant effect at all" (Schumpeter 2008, 80). Distinguishing between invention and innovation is another point that is really important from a philosophical point of view. Also, to understand how and why innovations are spread into society is a matter of concern for many companies, institutions and collectivities. This is something that is of vital importance in order to contribute to its popularization and social appropriation. In this sense, Everett Rogers (1962) made a great contribution in order to understand why some innovations are lucky and why others are not. Regarding the philosophy of science and technology, there is another important precaution: not all innovations come from scientific and technological knowledge, contrary to what the R + D + i cliché seems to assume. In a famous empirical based report, the British organization NESTA demonstrated that only 6–7% of the innovations produced in the United Kingdom had their origins in science (NESTA 2007, 5). Thus, there are many other sources of innovation apart from science and technology. Therefore, it is not prudent to try to extend the philosophy of technology to a philosophy of innovation. But at the same time we must face the problem of innovation from new conceptual perspectives, some of which could find inspiration in the philosophy of science and technology, while others will have to reflect on culture and the arts.[10] All of this increases the complexity of the task of creating a philosophy of innovation. One last precaution in relation to innovation studies lays on the different kinds of innovations that are characterized by The Oslo Manual. There are product innovations (new objects) but there are also innovations in processes, organization and marketing (or communications) processes. Since its 2005 version, the Oslo Manual distinguishes these four kinds of innovation in contrast to the 1992 version which was written based on the assumption that innovation consisted of generating new products (specially) in the manufacturing industry. We think that process innovations are more interesting because they are the ones that affect our human relations and interactions the most, just like organizational and communication innovations. For example; Internet has been a major innovation in communications, as well as a marketing innovation, independently of the new products transmitted or sold through the network. It also has proven to be a new way of organizing a community, a city or a society. Those kinds of innovations are also social innovations because there are relational innovations, not artifact innovations. The philosophy of innovation have to deal with new products and objects but at the same time it have to contribute to an understanding of innovation processes that are not in any way limited to these products and objects. We would like also to pay attention to user's innovation. The particularities of this new paradigm bring some light to the emerging paradigm of social innovation and especially in the case of ICT users/citizens mentioned earlier. In this sense, we could also specifically speak about a digital social innovation. A term that has been popularized in recent years and it has drawn the attention of some research initiatives in the European Union that they have highlighted its importance.[11] Eric von Hippel and his disciples were the ones who empirically demonstrated that in many economic subsectors, product suppliers, distributors and users are innovators. Not just manufacturers and producers (von Hippel 1988). These proposals hardly affected the innovation policies of the time but after the publication of "Democratizing Innovation" (2005), von Hippel's ideas have become canonical in innovation studies. The main idea is that, by using regularly a certain technological artifact (i.e. a bike, a text processor or a car) the user achieves a significant knowledge of the information product. That command allows users to detect faults and flaws that can be communicated to the manufacturer or distributor to improve the product and if appropriate, creating new innovative versions with suggestions provided by the users. This is also the case for users of services and it

[10] Cultural and creative sectors have proved to be highly innovative and the associated industries have gained a bigger impact on the GDP of different western economies.

[11] See for instance "Growing a Digital Ecosystem for Europe" at https://www.nesta.org.uk/sites/default/files/dsireport.pdf (Last retrieved on 09/09/1981).

affects all technological systems. Therefore, the knowledge that is useful for innovation comes not only from manufacturers and their R + D departments but from users. No matter if they are the final users or not. Other approaches in the sociology of technology likewise the “social shaping of technology” and the “domestication of technology” has pushed forward this kind of research (Baym 2015). Then, the service customer departments from many companies can become sources of innovation too if the users themselves are incorporated into the innovation processes. They can introduce improvements and modifications in the information objects that they use and provide new directions for the strategy of the company. From 2005 onwards, von Hippel’s theory has gained widespread acceptance because it has been endorsed by several empirical examples. As a result, the lineal model of innovation has been questioned and the OECD has begun to modify the concept of innovation (OECD 2009), just like the European Commission. The European Union Horizon 2020 strategy includes an entire emblematic initiative (Union Innovation 2010) destined to promote innovation in all of its forms called extended innovation. The aim is to support all kinds of innovations and not only those that are based on science and technology. It is generally accepted today that there are many sources of innovation as well as innovative agents. Among these, we can include NGOs, citizens’ associations, entities devoted to philanthropic works and some social movements. From this perspective, the emergence of unions, the fight for equal rights for women, Amnesty International, and Greenpeace are more relevant examples of social innovations though they were not based on R + D nor were they promoted by businesses.[12] By changing the concept of innovation and the notion of the innovating agent, new problems come out the philosophy of innovation, including also the convenience of reconsidering the concept of innovation itself. We argue that in innovation processes apart from contributing with something new (products, services, ways of organizing, marketing or communicating), a value must be generated but not only economic value. The dominant paradigm in innovation hold that statement but other kinds of value like social, cultural, ecologic, political and military are also noteworthy. That is the main reason why the sphere of innovation studies has to be considerably expanded. Embracing an axiological approach that considers innovation like a mix of values and accepts the notion of social value (CSI Stanford 2008) will be welcome. This proposal can be applied to information societies, where there are not only technological innovations but also social, legal, political and other kinds of innovations.

3.3 Dominant Values in the Infosphere

Before presenting the main lines of our axiological analysis of the third environment, we want to make it quite clear that when we speak of axiology, we are not referring to ethics but to the world of values in general as we have stressed at the beginning of the text. Ethics is focused on moral and religious values. We also

[12] This does not mean that they are not currently managing like businesses. Many social innovations end up generating institutions and companies.

want to emphasize that as Ortega y Gasset proposed in his Estimative in 1923, *the world of values cannot be reduced to ontology*, even though values and beings can compose among them. Identifying axiology with ethics is one of the great errors of 20th-century moral philosophy. Ortega was sensible enough not to make a mistake like this. He clearly stated that moral values are nothing more than one kind of value among many others (Ortega 1976, OC, VI, 334). We distinguish up to twelve kinds of values (Natural, Epistemic, Technological, Economic, Political, Military, Legal, Social, Ecological, Aesthetic, Religious, and Moral), most of which are relevant to technological issues and for the axiological analysis of the third environment. As a result, our axiological analysis is not oriented towards *Informational Ethics*, at least insofar as this discipline reduces axiology to ontology as is habitual.[13] We rather deal with the diverse axiological dimensions of information societies as well as their mixtures, interrelations and conflicts. Not just with one of them. We argue that moral values are not the dominant values either in technosocieties or in cybercities. This does not imply the absence of moral issues in the third environment. They are present and they are very important but morality focuses on private and intimate spheres. Nevertheless, the dominant values in the public spheres of information space are technological, economic, business, legal, epistemic, aesthetic and, in some cases, political and military values. These public spheres of information are dominated today by *information markets* (financial, media, sports, political, business, military markets, etc.). Moral values are rarely given priority and ethical values, either. Consider the stock markets interconnected by telematic networks to mention a canonical sphere of the third environment and make an empirical analysis of the values that govern the actions that the financial agents carry out there. This is the methodology for analyzing any sphere of the third environment in an axiological way and not postulating what should be done without even analyzing what is happening. We do not deny that some information objects can have a moral value, as Floridi states. But what happens usually is that some innovative digital objects acquire a high economic value as well as technological, social, political or a military one. Just to mention five kinds of value that are almost always relevant in the third environment. After having made these points, we would like to briefly mention some specific values. First of all, technological values such as applicability, utility, rapidity, robustness, capacity (for storage, for transmission), usability, trustworthiness, etc.; without forgetting efficiency or efficacy are highly relevant in the third environment. Efficiency and efficacy are also technological values per se for Quintanilla (1989). We will not provide specific examples here because there is no need. Any ICT user could easily think about several examples after reading this first listing of technological values. This list, as a matter of fact, is open and not exhaustive. It can be easily extended as there are many technological values. In second place, we must not forget military values. There were very present in the origin of Internet (Arpanet was the origin of Internet but also of Milnet) and they continue to be decisive in broad areas of the third environment. These areas have

[13] Luciano Floridi, for example, when he discusses the axiology of information objects only refers to moral issues. When he accepts the Kantian notion of intrinsic value he goes so far as to state that: “The moral value of an entity is based on its ontology. What the entity is determines the degree of moral value it enjoys” (L. Floridi 2002, p. 20).

been growing in recent decades even though no one talks about it. Drones are an example of a digital information object. They usually are used in bombings by military agencies but lately civic uses have appeared. This has happened often throughout the evolution of the third environment; military innovations end up generating innovations for civil use. Military information actions tend to be guided by values such as security, defense, victory, secret, discipline, due obedience, maximizing harm to the adversary, etc. Moral values are not very present in these spheres, at least from a Kantian point of view in which human beings are ends in themselves. And a military IT object too. The third group of values that need to be discussed are political values. In this regard, it seems clear that the struggle to conquer political power in the States has become a political techno-market. The use of the old media (press, radio, television) and the new media (Internet, social media) has become a fundamental part of the political battle as many authors have shown. Particularly, Castells (2011) is one of the biggest contributors. Luckily, in this struggle for political power not everything is acceptable and there are still legal systems that maintain values such as equality, freedom, solidarity, justice and legality. But the tensions around these values and the agents who uphold them (legislators, judges) is increasing. On behalf of innovation, there are many who desire an unregulated third environment or a techno-regulated one, where the technological and business agents themselves would be the ones to establish the rules of the game. The intervention of the States, according to many of the ideologists of information capitalism, should be minimal and subsidiary to the technoscientific power dominant today. This problem has been addressed by many authors in the field of ICT (Terranova 2004; Fuchs 2010; Scholz 2012) and some of them have coined the term "Californian Ideology" (Barbrook and Cameron 1996) in order to express how the new Big Players from Silicon Valley are imposing their ideology and values into different societies across the globe. Expressing a new kind of liberalism mixed with some of the values of the contra-culture that was rapidly spread in the 70's. Last but not least, economic values must be mentioned. These values are generally the main ones in the third environment and in the infosphere. A financial product (i.e. a *sub-prime* mortgage) is a typical information object whose existence in the third environment deserves a precise axiological analysis. We can say the same about a television commercial at midnight slot, a message on Twitter that becomes a "trending topic" or about the worldwide audience of a big sports event. The "intrinsic value" of these kinds of objects is mostly an economic value. Even though, in a subsidiary way they can have a certain moral value, positive or negative. What we want to highlight is that the moral values are clearly subordinated to the economic values in the vast majority of the information objects of the third environment. This is similar to the way they were subordinated to the political, military, and technological values. If we analyze Kantian categorical imperatives from the perspective of technological values, we can see that there are not at all efficient. In fact, hardly anyone follows them in the real world and this is why they have been relegated to the world of how things should be, with no effectiveness in information society or in the processes of innovation.

3.4 Res publica in the Third Environment?

The hypothesis that we want to stress is that there won't be *res publica* in the infosphere until public institutions create, promote and maintain a public space in the third environment. This info-agora would be based on free access to all the info-objects existing there. These implies open access to information and knowledge and what means public goods not only community or social goods. Based on this general hypothesis, we will briefly evoke the utopia of a democratic Telepolis in which *info-objects are freely shared*. This is something that has been argued by other authors (Rifkin 2014; Bauwens and Kostakis 2015). These scholars have praised the potential of the collaborative economy and other alternatives to the current decline of capitalism. This project still seems possible to us, although there are great obstacles to progress due to present-day relations of power, subordination and property in the third environment. It seems that there are not many relevant techno-scientific agents in favor of a democratic cybercity. Just some communities of users and hacktivists have expressed concern and awareness about it. These are the ones who understand the infosphere as a civil information space. Not only as a technological, business, military and financial one. The most powerful technological agents (the Lords of the Air or the Lords of the Networks, in our terminology) have everything they need with the Cybermarkets. They do not seem to consider the usefulness of a global Cybercity with an Info-agora and a system of governance that is open to the users. To sum up, Telepolis has an aristocratic government structure. The eventual emergence of cybercitizens from *info-users* would radically modify relations in the third environment and would mean an important social innovation.

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